serviceOrchestrationLockManagement MQTT/TLS/JSON

Interface Design Description

Service ID: *"serviceOrchestrationLockManagement"*

Abstract
This document describes the MQTT protocol with TLS payload security and JSON payload encoding variant of the `serviceOrchestrationLockManagement` service of the DynamicServiceOrchestration system.

Contents

1 Documentation edition and implementation scope **3**

2 Overview **3**

3 Service Operations **4**
3.1 operation create . . . 4
3.2 operation query . . . 5
3.3 operation remove . . . 5

4 Data Models **6**
4.1 struct MqttRequestTemplate . . . 6
4.2 struct MqttResponseTemplate . . . 6
4.3 struct ErrorMessage . . . 6
4.4 struct OrchestrationLockListRequest . . . 6
4.5 struct OrchestrationLockListResponse . . . 6
4.6 struct OrchestrationLockQueryRequest . . . 6
4.7 struct OrchestrationLockRequest . . . 7
4.8 struct OrchestrationLockResponse . . . 7
4.9 struct Page . . . 7
4.10 Primitives . . . 8

5 Go and Rust implementation variants **9**
5.1 Go implementation . . . 9
5.2 Rust implementation . . . 9

6 References **9**

7 Revision History **10**
7.1 Amendments . . . 10
7.2 Quality Assurance . . . 10

1 Documentation edition and implementation scope

This is the 5.3.0 documentation edition, dated 2026-11-09. It carries forward the system and service contracts from the retained 5.2.2 documentation and identifies the current primary implementation sources. The former `documentation/v5.2.2` folder is retained unchanged.

Document version 5.3.0 is distinct from executable, dependency, registered system/service and wire-contract versions. Values still defined by the source (including 5.2.2 artefact versions, 1.0.0 service versions and `SysMonitor-AMQP-IDD-5.2.2` payload identifiers) are preserved. This documentation update does not change runtime behaviour or establish a new implementation release. Historical verification statements describe the tests recorded in the inherited baseline; this revision validates document compilation and references.

2 Overview

This document describes the `serviceOrchestrationLockManagement` service interface of the DynamicServiceOrchestration system over MQTT, which provides operator management of orchestration locks, the exclusive reservations of service instances.

It's implemented using protocol, encoding as stated in below table:

Profile type	Type	Version
Transfer protocol	MQTT	3.1.1 [1]
Data encryption	TLS	1.3 [2], 1.2 [3]
Encoding	JSON	RFC 8259 [4]
Compression	N/A	-
Semantics	N/A	-
Ontology	N/A	-

Table 1: Communication and semantics details used for the `serviceOrchestrationLockManagement` service interface

This document provides the Interface Design Description IDD to the *serviceOrchestrationLockManagement – Service Description* document [5]. The operations, their semantics and the data models are the same as in the HTTP/TLS/JSON IDD [6]; this document specifies the MQTT binding.

The MQTT interface is offered only when the system runs with `mqtt.api.enabled=true`; it is then registered with the interface template `generic_mqtts` (`generic_mqtt` without TLS), the broker address and port (`mqtt.broker.address`, `mqtt.broker.port`) and the base topic `arrowhead/serviceorchestration/orchestration/management/lock/`. In the Arrowhead-test-cloud-java deployment the MQTT interfaces of the core systems are disabled.

3 Service Operations

The provider subscribes to one topic per operation, the base topic followed by the operation name:

Operation	Request topic
`create`	`arrowhead/serviceorchestration/orchestration/management/lock/create`
`query`	`arrowhead/serviceorchestration/orchestration/management/lock/query`
`remove`	`arrowhead/serviceorchestration/orchestration/management/lock/remove`

A consumer publishes a request envelope (MqttRequestTemplate) to the topic and receives the answer (MqttResponseTemplate) on its own `responseTopic`. The `authentication` field carries the requester's identity according to the provider's authentication policy: `IDENTITY-TOKEN//<token>` (outsourced or internal policy), `SYSTEM//<name>` (declared policy) or the requester's X.509 certificate (certificate policy). The `status` of the response uses the HTTP status code numbers (200, 201, 204, 400, 401, 403, 404, 408, 423, 500, 503); on error the payload is an ErrorMessage.

```
{
  "traceId": "a1b2c3",
  "authentication": "IDENTITY-TOKEN//<token>",
  "responseTopic": "arrowhead/response/TemperatureDisplay/a1b2c3",
  "qosRequirement": 1,
  "params": {},
  "payload": {}
}
```

Listing 1: Request envelope.

```
{
  "status": 200,
  "traceId": "a1b2c3",
  "receiver": "TemperatureDisplay",
  "payload": {}
}
```

Listing 2: Response envelope.

3.1 PUBLISH `arrowhead/serviceorchestration/orchestration/management/lock/create`

Operation: create
Input: OrchestrationLockListRequest
Output: OrchestrationLockListResponse

Creates locks on service instances for owners.

```
{
  "locks": [
    {
      "serviceInstanceId": "TemperatureSensor|temperatureReading|1.0.0",
      "owner": "TemperatureDisplay",
      "expiresAt": "2026-10-22T12:00:00Z"
    }
  ]
}
```

Listing 3: Payload of an create request.

3.2 PUBLISH arrowhead/serviceorchestration/orchestration/management/lock/query

Operation: query
Input: OrchestrationLockQueryRequest
Output: OrchestrationLockListResponse

Returns a page of locks matching the filter.

```
{
  "pagination": {
    "page": 0,
    "size": 10,
    "direction": "ASC",
    "sortField": "createdAt"
  },
  "ids": [
    1
  ],
  "orchestrationJobIds": [
    "3fa85f64-5717-4562-b3fc-2c963f66afa6"
  ],
  "serviceInstanceIds": [
    "TemperatureSensor|temperatureReading|1.0.0"
  ],
  "owners": [
    "TemperatureDisplay"
  ],
  "expiresBefore": "2026-10-22T12:00:00Z",
  "expiresAfter": "2026-10-22T12:00:00Z"
}
```

Listing 4: Payload of an query request.

3.3 PUBLISH arrowhead/serviceorchestration/orchestration/management/lock/remove

Operation: remove
Input: Array<String>

Removes the locks of an owner on the given service instances.

```
[
  "TemperatureDisplay"
]
```

Listing 5: Payload of an remove request.

4 Data Models

Here, all data objects that can be part of the service calls associated with this service are listed in alphabetic order. Each *struct* denotes a JSON Object with the named fields; fields that are not needed may be omitted.

4.1 struct MqttRequestTemplate

The request envelope.

Object Field	Value Type	Description
"traceId"	String	Identifier chosen by the requester, returned in the response.
"authentication"	String	Identity of the requester (see Section 3).
"responseTopic"	String	Topic on which the response is published.
"qosRequirement"	Number	MQTT QoS of the response (0, 1 or 2).
"params"	Object<String>	Optional parameters (the query parameters of the HTTP binding).
"payload"	Object	The operation input.

4.2 struct MqttResponseTemplate

The response envelope.

Object Field	Value Type	Description
"status"	Number	Status code.
"traceId"	String	The trace identifier of the request.
"receiver"	String	Name of the requester.
"payload"	Object	The operation output, or an ErrorMessage.

4.3 struct ErrorMessage

Object Field	Value Type	Description
"errorMessage"	String	
"errorCode"	Number	
"exceptionType"	ExceptionType	
"origin"	String	

4.4 struct OrchestrationLockListRequest

Locks to create.

Object Field	Value Type	Description
"locks"	Array<OrchestrationLockRequest>	Locks to create.

4.5 struct OrchestrationLockListResponse

List of orchestration locks.

Object Field	Value Type	Description
"entries"	Array<OrchestrationLockResponse>	Matching records of the current page.
"count"	Number	Total number of matching records (all pages).

4.6 struct OrchestrationLockQueryRequest

Paged filter for orchestration locks.

Object Field	Value Type	Description
"pagination"	Page	Paging and sorting of the result.
"ids"	Array<Number>	Lock identifiers to match.
"orchestrationJobIds"	Array<String>	Orchestration job identifiers to match.
"serviceInstanceIds"	Array<String>	Service instance identifiers to match.
"owners"	Array<String>	Lock owner names to match.
"expiresBefore"	DateTime	Upper bound of the expiry time.
"expiresAfter"	DateTime	Lower bound of the expiry time.

4.7 struct OrchestrationLockRequest

Exclusive reservation of a service instance.

Object Field	Value Type	Description
"serviceInstanceId"	String	Service instance identifier `<System>|<serviceDefinition>|<version>`.
"owner"	String	System for which the instance is locked.
"expiresAt"	DateTime	UTC expiry of the lock.

4.8 struct OrchestrationLockResponse

A lock on a service instance.

Object Field	Value Type	Description
"id"	Number	Identifier of the lock.
"orchestrationJobId"	String	Orchestration job that created the lock.
"serviceInstanceId"	String	Service instance identifier `<System>|<serviceDefinition>|<version>`.
"owner"	String	Name of the system owning the lock.
"expiresAt"	DateTime	UTC expiry of the lock.
"temporary"	Boolean	True for a lock held only during an orchestration.

4.9 struct Page

Paging and sorting; all fields optional.

Object Field	Value Type	Description
"page"	Number	Zero-based page number.
"size"	Number	Number of records per page (at most `max.page.size`, default 1000).
"direction"	String	Sort direction, `ASC` or `DESC`.
"sortField"	String	Field to sort by.

4.10 Primitives

As all messages are encoded using the JSON format [4], the following primitive constructs, part of that standard, become available.

Type	Description
Array	A JSON Array [4]; Array<A> has elements of type A.
Boolean	JSON Boolean, `true` or `false`.
DateTime	JSON String with an ISO 8601 UTC timestamp, e.g. `2026-10-22T12:00:00Z`.
ExceptionType	One of `ARROWHEAD`, `INVALID_PARAMETER`, `AUTH`, `FORBIDDEN`, `DATA_NOT_FOUND`, `TIMEOUT`, `LOCKED`, `INTERNAL_SERVER_ERROR`, `EXTERNAL_SERVER_ERROR` (JSON String).
Number	JSON Number, an integer.
Object	JSON Object; Object<A> has string keys and values of type A.
String	JSON String.

5 Go and Rust implementation variants

5.1 Go implementation

Primary system folder: `Eclipse-Arrowhead-go/ah5-core-go/serviceorchestration-dynamic`.

The system owns its Go module, launcher and build configuration. Use the per-system `Makefile` and `go.mod`; the containing Go tree is the build source, not a historical Arrowhead copy. Ordinary native roles use independent identities and JSON snapshot state rather than the Java database. Do not assume Java SQL migration, clustering or identical transaction/retry semantics. The registry has its own configuration and optional native AMQP monitor; other roles must be checked against their own registrations.

Listening port. Built-in default 127.0.0.1:8441 (Arrowhead default). The shipped config.json sets "listen": "0.0.0.0:8441". Override with "listen" in the configuration file, SERVER_PORT=N or -server.port N; the port overrides keep the configured host. Native design/route evidence: `Eclipse-Arrowhead-go/ah5-core-go/serviceorchestration-dynamic/documentation`.

For this Java interface, select only a native transport and operation explicitly implemented and advertised by that port. A matching method/path does not establish DTO, error, authorization, retry or persistence equivalence. Java MQTT/AMQP bindings are not automatically provided by a Go or Rust role. Differential contract and security checks remain necessary before substitution.

5.2 Rust implementation

Primary system folder: `Eclipse-Arrowhead-rust/ah5-core-rust/serviceorchestration-dynamic`.

The per-system entry point is built by the `arrowhead-core-rust` crate. Build from the primary Rust workspace with the applicable package/binary selection. Documentation links can resolve into the shared core/support catalogue. Independent process identity and state remain required.

The ordinary native profile uses Tokio/Axum and rustls with peer-identity checks and independent locked JSON state. Network effects are not distributed database transactions. The precise trust, CRL reload and persistence guarantees are those of the native profile, not the Java implementation.

Native design/route evidence: `Eclipse-Arrowhead-rust/ah5-core-rust/documentation/DynamicServiceOrchestration`.

For this Java interface, select only a native transport and operation explicitly implemented and advertised by that port. A matching method/path does not establish DTO, error, authorization, retry or persistence equivalence. Java MQTT/AMQP bindings are not automatically provided by a Go or Rust role. Differential contract and security checks remain necessary before substitution.

6 References

[1] A. Banks and R. Gupta, “Mqtt version 3.1.1,” OASIS, Standard, 2014.

[2] E. Rescorla, “The transport layer security (tls) protocol version 1.3,” IETF, RFC 8446, 2018.

[3] T. Dierks and E. Rescorla, “The transport layer security (tls) protocol version 1.2,” IETF, RFC 5246, 2008.

[4] T. Bray, “The javascript object notation (json) data interchange format,” IETF, RFC 8259, 2017.

[5] J. Delsing, “serviceOrchestrationLockManagement – Service Description (DynamicServiceOrchestration),” Eclipse Arrowhead, SD 5.3.0, 2026, dynamicServiceOrchestration-v5.3.0-serviceOrchestrationLockManagement-SD.pdf.

[6] ——, “serviceOrchestrationLockManagement HTTP/TLS/JSON – Interface Design Description (DynamicServiceOrchestration),” Eclipse Arrowhead, IDD 5.3.0, 2026, dynamicServiceOrchestration-v5.3.0-serviceOrchestrationLockManagement-HTTP-IDD.pdf.

7 Revision History

7.1 Amendments

No.	Date	Version	Subject of Amendments	Author
1	2026-10-22	5.2.2	Document created from the Arrowhead LaTeX templates for the 5.2.2 Java Spring implementation, including the local extensions of this workspace.	Jerker Delsing
2	2026-11-09	5.3.0	New documentation edition; primary-source references and Go/Rust implementation scope; 5.2.2 retained.	Documentation maintenance

7.2 Quality Assurance

No.	Date	Version	Approved by
1		5.3.0	